Cristobal Cortés Gómez

Software Developer & Mathematician

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Professional Profile

Cross-platform developer across desktop, mobile, web and embedded systems — mostly game development, with the most recent projects built on web technologies. Experienced in porting, extending and debugging existing codebases. Mathematics graduate with a background in machine learning; advanced English, used daily with U.S.-based clients.

Technical Skills

Programming languages: TypeScript/JavaScript, Python, C, C#, GML, GLSL.
Web, frameworks & tools: HTML/CSS, Node.js, React, Redux, Three.js, Firebase, GameMaker, Unity 3D.
Systems & hardware: Linux (shell scripting), Git, Emacs, Vim; ESP8266, Arduino, PLC.
Spoken languages: Spanish (native); English (advanced).

Experience

2025 – Jun. 2026: Software Developer — *Gnarled Helix LLC (U.S. studio), remote contract*

- Developed a 2D action game that teaches programming (TypeScript, Three.js, React, Redux, Node.js), built around a runtime that executes player-written JavaScript to drive the character's abilities.
- Worked within a multidisciplinary, distributed team on an existing codebase.

2024–2025: Designer & Sole Developer — *Jalapeño Labs, contract*

- Led the video game adaptation of the board game "Camino a Xibalbá PITZ" (GameMaker, JavaScript, Firebase), translating a printed rule set into working game logic and checking behavior against those rules case by case.
- Owned the correctness of turn and rule resolution, and the Firebase data layer behind player accounts and match records.

2023–2024: Embedded Systems & Automation — *Private client, contract*
Built an automated greenhouse for edible mushroom cultivation, with humidity and temperature control: firmware, PCB design and the complete electrical installation (ESP8266, Arduino IoT).

2022–2024: Mathematics Tutor — *Independent*
2022: Robotics Instructor — *American World Languages*

2020–2022: Hardware & Software Developer — *Proyecto INTERSEX*
Built an LED face mask that displays custom images, and a 3D museum simulator, for a social and artistic project (JavaScript, Arduino, ESP8266, Blender, Unity 3D).

2017–2020: Web Developer — *SGT Total*
Built a mobile-responsive commercial website over several months, then handled occasional change requests from the client in the years that followed.

2017–2018: Freelance projects — *Upwork and direct clients*
Immertec (2018, Upwork): ported a GLSL shader to HLSL, delivering a spherical/fisheye VR projection for a medical application. *Daniel Douglas* (2018, Upwork): desktop-simulation mini-game built to a written client brief. *"Habitar" theater production* (2017): Kinect application adjusting real-time projections to physical set movement (Processing, Java, GLSL).

2016–2017: Industrial Systems & Automation — *Cervejal*
Built a semi-automatic bottle labeling machine and programmed a Siemens PLC (ladder logic) for a keg washer/filler, commissioned and tested on site.

Selected Independent Projects

Shipped games: "Morbus" (2012), "Sweet Valentine's Day" (2014), "Sacrifice Your Brothers" (2018), "Me Canso Ganso" (2019).
Other game projects: "Gravity Box" (2018, unreleased); "Libélula" (in development).

Education

2025: Bachelor's Degree in Mathematics — Universidad de Guadalajara (CUCEI). Final project: "Suitability Analysis of Shallow vs. Deep Neural Networks Using the Perceptron for Classification Problems".